Exact Norm of an Unresolved Lorentz Sequence Embedding

Candidate solution packet for arXiv:2510.24290

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Status. Candidate full solution, likely valid, pending human review.

1 Source question

Kneselová studies the Lorentz sequence quasi-norms

$$\|a\|_{p,q} = \left(\sum_{n=1}^{\infty} n^{q/p-1}(a_n^*)^q\right)^{1/q} \qquad (q<\infty), \qquad \|a\|_{p,\infty} = \sup_{n\geq 1} n^{1/p}a_n^*, \tag{1}$$

where a^* is the decreasing rearrangement of $(|a_n|)$ [1]. For

$$0 < p_1 < p_2 \leq \infty, \qquad p_1 < q_1 < q_2 \leq \infty,$$

Theorem 3.4(3) proves

$$\|I : \ell^{p_1,q_1} \to \ell^{p_2,q_2}\| \leq \left(\frac{q_1}{p_1}\right)^{1/q_1-1/q_2}, \tag{2}$$

with $1/q_2 = 0$ at $q_2 = \infty$. On PDF page 15 the source states that it is not known whether this constant is optimal in full generality, apart from the diagonal endpoint treated in Theorem 3.6.

2 Proof intuition

The decreasing cone has very simple generators. If $z_1 \geq z_2 \geq \cdots \geq 0$ and $z_n \to 0$, then

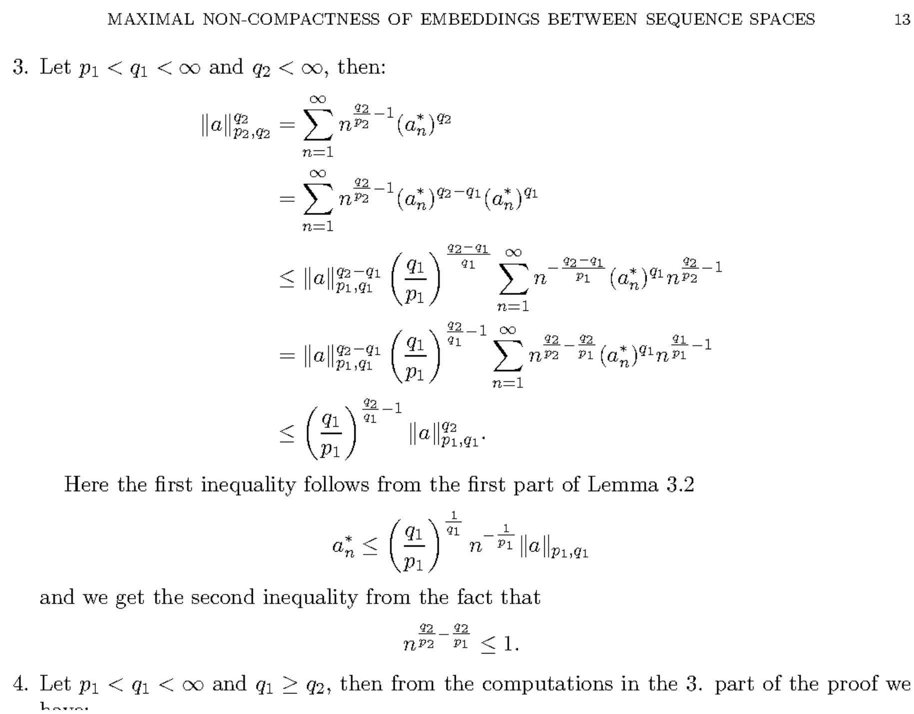
$$z = \sum_{N\geq 1}(z_N - z_{N+1})\mathbf{1}_{[1,N]}.$$

After setting $z_n = (a_n^*)^{q_1}$, the source modular is linear in z, whereas the target modular becomes a weighted ℓ_r norm with $r = q_2/q_1 > 1$. Minkowski's inequality says that a positive sum of the initial-segment generators cannot have a larger norm-to-budget ratio than the worst generator. Consequently initial-segment indicators determine the exact operator norm.

3 Exact norm

For $N \in \mathbb{N}$ set

$$B_N = \sum_{n=1}^{N} n^{q_1/p_1-1}. \tag{3}$$



3. Let $p_1 < q_1 < \infty$ and $q_2 < \infty$, then:

$$\begin{aligned}
\|a\|_{p_2,q_2}^{q_2} &= \sum_{n=1}^{\infty} n^{\frac{q_2}{p_2}-1} (a_n^*)^{q_2} \\
&= \sum_{n=1}^{\infty} n^{\frac{q_2}{p_2}-1} (a_n^*)^{q_2-q_1} (a_n^*)^{q_1} \\
&\le \|a\|_{p_1,q_1}^{q_2-q_1} \left(\frac{q_1}{p_1}\right)^{\frac{q_2-q_1}{q_1}} \sum_{n=1}^{\infty} n^{-\frac{q_2-q_1}{p_1}} (a_n^*)^{q_1} n^{\frac{q_2}{p_2}-1} \\
&= \|a\|_{p_1,q_1}^{q_2-q_1} \left(\frac{q_1}{p_1}\right)^{\frac{q_2}{q_1}-1} \sum_{n=1}^{\infty} n^{\frac{q_2}{p_2}-\frac{q_2}{p_1}} (a_n^*)^{q_1} n^{\frac{q_1}{p_1}-1} \\
&\le \left(\frac{q_1}{p_1}\right)^{\frac{q_2}{q_1}-1} \|a\|_{p_1,q_1}^{q_2}.
\end{aligned}$$

Here the first inequality follows from the first part of Lemma 3.2

$$a_n^* \le \left(\frac{q_1}{p_1}\right)^{\frac{1}{q_1}} n^{-\frac{1}{p_1}} \|a\|_{p_1,q_1}$$

and we get the second inequality from the fact that

$$n^{\frac{q_2}{p_2}-\frac{q_2}{p_1}} \le 1.$$

4. Let $p_1 < q_1 < \infty$ and $q_1 \ge q_2$, then from the computations in the 3. part of the proof we

Figure 1: Theorem 3.4(3) and the source estimate on PDF page 13.

When $q_2 < \infty$, also set

$$A_N = \sum_{n=1}^{N} n^{q_2/p_2-1}, \tag{4}$$

using $q_2/p_2 = 0$ if $p_2 = \infty$. We use the convention $N^{1/p_2} = 1$ if $p_2 = \infty$.

Theorem 1. *Let $0 < p_1 < p_2 \le \infty$ and $p_1 < q_1 < q_2 \le \infty$. Then*

$$\|I : \ell^{p_1,q_1} \to \ell^{p_2,q_2}\| = \max_{N\ge1} \frac{A_N^{1/q_2}}{B_N^{1/q_1}}, \qquad q_2 < \infty, \tag{5}$$

$$\|I : \ell^{p_1,q_1} \to \ell^{p_2,\infty}\| = \max_{N\ge1} \frac{N^{1/p_2}}{B_N^{1/q_1}}. \tag{6}$$

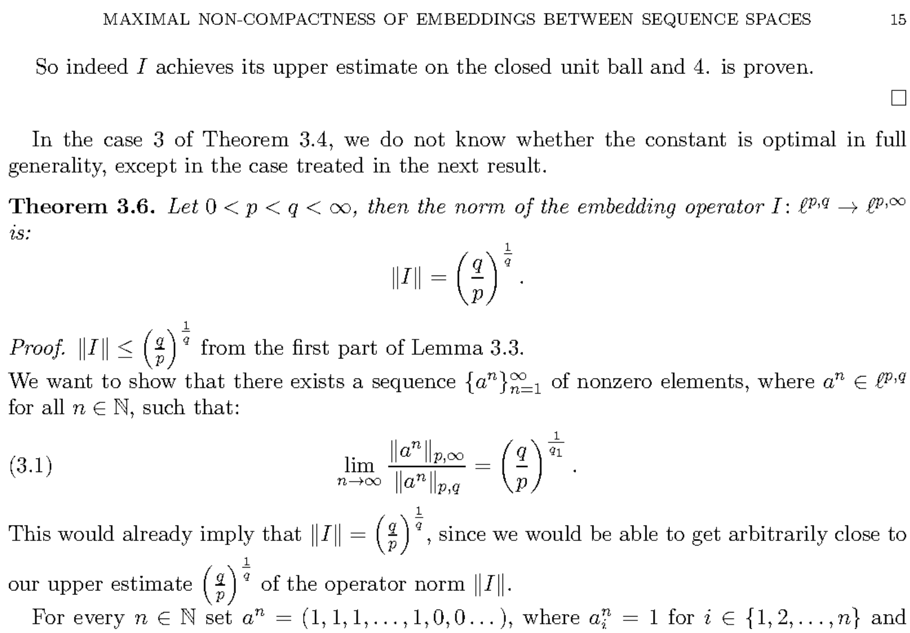
In both formulas a maximizing vector is an initial-segment indicator. Moreover,

$$\|I\| < \left(\frac{q_1}{p_1}\right)^{1/q_1-1/q_2}. \tag{7}$$

Thus the constant in (2) is never optimal in the off-diagonal range $p_1 < p_2$.

Proof. It is enough to work with $x = a^*$, so x is nonnegative and decreasing. First suppose $q_2 < \infty$. Put

$$r = \frac{q_2}{q_1} > 1, \qquad z_n = x_n^{q_1}, \qquad d_N = z_N - z_{N+1} \ge 0.$$


MAXIMAL NON-COMPACTNESS OF EMBEDDINGS BETWEEN SEQUENCE SPACES 15

So indeed I achieves its upper estimate on the closed unit ball and 4. is proven. $\square$

In the case 3 of Theorem 3.4, we do not know whether the constant is optimal in full generality, except in the case treated in the next result.

Theorem 3.6. *Let $0 < p < q < \infty$, then the norm of the embedding operator $I\colon \ell^{p,q} \to \ell^{p,\infty}$ is:*

$$\|I\| = \left(\frac{q}{p}\right)^{\frac{1}{q}}.$$

Proof. $\|I\| \le \left(\frac{q}{p}\right)^{\frac{1}{q}}$ from the first part of Lemma 3.3.
We want to show that there exists a sequence $\{a^n\}_{n=1}^{\infty}$ of nonzero elements, where $a^n \in \ell^{p,q}$ for all $n \in \mathbb{N}$, such that:

$$\lim_{n\to\infty} \frac{\|a^n\|_{p,\infty}}{\|a^n\|_{p,q}} = \left(\frac{q}{p}\right)^{\frac{1}{q_1}}. \tag{3.1}$$

This would already imply that $\|I\| = \left(\frac{q}{p}\right)^{\frac{1}{q}}$, since we would be able to get arbitrarily close to our upper estimate $\left(\frac{q}{p}\right)^{\frac{1}{q}}$ of the operator norm $\|I\|$.
For every $n \in \mathbb{N}$ set $a^n = (1,1,1,\dots,1,0,0\dots)$, where $a_i^n = 1$ for $i \in \{1,2,\dots,n\}$ and

Figure 2: The explicit open sentence on PDF page 15.

The finiteness of the source quasi-norm implies $z_n \to 0$. Hence, pointwise,

$$z = \sum_{N=1}^{\infty} d_N \mathbf{1}_{[1,N]}. \tag{8}$$

Let $v_n = n^{q_2/p_2-1}$ and $w_n = n^{q_1/p_1-1}$, and write

$$\|u\|_{r,v} = \left(\sum_{n\ge 1} v_n |u_n|^r\right)^{1/r}.$$

For a finite partial sum in (8), Minkowski's inequality gives

$$\left\|\sum_{N=1}^{M} d_N \mathbf{1}_{[1,N]}\right\|_{r,v} \le \sum_{N=1}^{M} d_N A_N^{1/r}.$$

Both sides increase to their infinite counterparts. Weighted monotone convergence on the left and Tonelli's theorem on the right therefore give

$$\begin{aligned}
\|z\|_{r,v} &\le \sum_{N\ge 1} d_N A_N^{1/r} \\
&\le \left(\sup_{N\ge 1} \frac{A_N^{1/r}}{B_N}\right) \sum_{N\ge 1} d_N B_N \\
&= \left(\sup_{N\ge 1} \frac{A_N^{1/r}}{B_N}\right) \sum_{n\ge 1} w_n z_n.
\end{aligned}$$

The last equality follows because $\sum_{N\geq n} d_N = z_n$. In terms of x, this is

$$\|x\|_{p_2,q_2}^{q_1} \leq \left(\sup_{N\geq 1} \frac{A_N^{q_1/q_2}}{B_N}\right) \|x\|_{p_1,q_1}^{q_1}.$$

Taking q_1-th roots gives the upper bound in (5). For $x = \mathbf{1}_{[1,N]}$, the norm ratio is exactly $A_N^{1/q_2}/B_N^{1/q_1}$, proving equality.

Now suppose $q_2 = \infty$. Since x is decreasing,

$$B_N x_N^{q_1} \leq \sum_{n=1}^{N} w_n x_n^{q_1} \leq \|x\|_{p_1,q_1}^{q_1}.$$

Consequently

$$N^{1/p_2} x_N \leq \frac{N^{1/p_2}}{B_N^{1/q_1}} \|x\|_{p_1,q_1}.$$

Taking the supremum over N proves the upper bound in (6), and $x = \mathbf{1}_{[1,N]}$ again gives equality.

It remains to justify the maxima and strictness. If $q_2 < \infty$ and $p_2 < \infty$, the power-sum asymptotics give

$$\frac{A_N^{1/q_2}}{B_N^{1/q_1}} \sim \left(\frac{p_2}{q_2}\right)^{1/q_2} \left(\frac{q_1}{p_1}\right)^{1/q_1} N^{1/p_2-1/p_1} \longrightarrow 0.$$

If $p_2 = \infty$, A_N is the N-th harmonic sum and the same ratio is $O((\log N)^{1/q_2} N^{-1/p_1})$, hence again tends to zero. The ratios in (6) tend to zero by the same power-sum estimate (also when $p_2 = \infty$). Thus every displayed supremum is attained.

Let

$$C = \left(\frac{q_1}{p_1}\right)^{1/q_1-1/q_2} > 1.$$

For finite q_2, repeat the source estimate on $x = \mathbf{1}_{[1,N]}$. When $N \geq 2$, its last step is strict because $n^{q_2(1/p_2-1/p_1)} < 1$ for every $n \geq 2$ and all summands are positive. Hence $A_N^{1/q_2}/B_N^{1/q_1} < C$. For $N = 1$, the ratio is $1 < C$. When $q_2 = \infty$, the strict right Riemann-sum estimate

$$B_N = \sum_{n=1}^{N} n^{q_1/p_1-1} > \frac{p_1}{q_1} N^{q_1/p_1}$$

gives

$$\frac{N^{1/p_2}}{B_N^{1/q_1}} < \left(\frac{q_1}{p_1}\right)^{1/q_1} N^{1/p_2-1/p_1} \leq C.$$

Since the supremum is attained at a finite N, these pointwise strict inequalities imply (7). □

4 Verification and limitations

The script `code/verify_exact_norm.py` sampled 4,000 decreasing random-jump sequences in each of five finite-dimensional parameter regimes, including $q_1 < 1$, $q_2 = \infty$, and $p_2 = \infty$. Every observed ratio was bounded by the appropriate initial-block maximum, and a maximizing block attained it to floating-point tolerance. This is a sanity check, not part of the proof.

The run's lightweight indexes contain no duplicate for arXiv:2510.24290 or the exact norm question. A bounded external search used the exact title and question together with "Lorentz sequence space", "exact norm", "embedding", and weighted inequalities on decreasing sequences. No separate paper applying this formula to the extracted question was located. General weighted inequalities on monotone cones are classical, so absolute novelty is provisional pending specialist review; the argument above is self-contained.

Human review recommendation. Check the passage from finite jump sums to (8), the exponent bookkeeping before (5), and the blockwise strictness at the end. No conditional lemma remains.

References

[1] A. Kneselová, *Maximal non-compactness of embeddings between sequence spaces*, arXiv:2510.24290, 2025, especially Theorem 3.4(3) and p. 15.